Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-7 | Winter-2024

Course Code	:	2101CS701	**Date**	:	19-10-2024
Course Name	:	Compiler Design	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 (A) Explain language translation process. **4**

(B) Explain types of compiler. **3**

OR

Explain linker and loader.

(C) Explain phases of compiler. **7**

OR

Differentiate interpreter and compiler.

Q.2 (A) Explain input buffering method with suitable example. **4**

(B) Differentiate DFA and NFA. **3**

OR

Draw a DFA for binary string ends with 010.

(C) Convert the following RE into DFA using Thomson's construction: (a + b)* b **7**

OR

Convert the following RE into DFA: a*b (a + b)* ab

Q.3 (A) How do you check whether the grammar is LL (1) or not? Generate LL(1) parsing table for given Grammar **4**
E→E+T | T
T→T*F | F
F→id | (E)
Is Grammar LL(1) or not?

(B) Differentiate top down parsing and bottom up parsing. **3**

OR

Explain shift reduce parser.

(C) Construct SLR parsing table for the following grammar: **7**
D→TL;
T→ i | f
L→L,id | id

OR

Explain synthesized attribute with suitable example.

Q.4	**(A)**	Write a different forms of Intermediate code generation for given input string: X=(a+b)*(c+d)+(a+b+c)	**4**
	(B)	Explain activation record.	**3**

OR

Explain various parameter passing methods.

	(C)	Explain Quadruples, Triples & Indirect Triples With example.	**7**

OR

Explain dynamic storage allocation.

Q.5	**(A)**	Explain Peephole Optimization in detail.	**4**
	(B)	Explain basic block and flow graph.	**3**

OR

What is Symbol table? Discuss different data Structures suitable for it.

	(C)	Discuss generic issues in the design of code generation	**7**

OR

Explain various code optimization techniques.

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